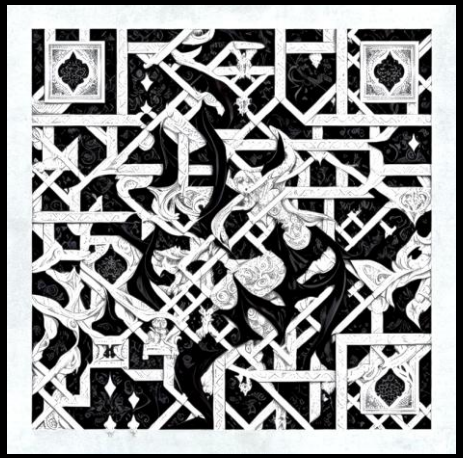
DEVFEST
NANTES . 2025



Embellir des QR Codes à l'ère de la GenAI

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WORLDLINE



Raphaël Semeteys

Head of DevRel @Worldline
Architecte Sénior
Expert Open Source

QR Codes

Quick Response Codes

Origines

- Conçus en 1994 chez Toyota subsidiary
- Standards ISO/IEC 18004:2000/2006/2015/2024
- Inspirés du jeu de Go

Evolution 2D des code-barres

- Motifs carrés noirs sur fond blanc
- Contenu (alpha)numérique, binaire ou Kanji

Différentes tailles (ou versions)

- De v1 (21x21) à v40 (177x177)
- Chaque version est 4 fois plus grande

Mécanisme intégré de correction d'erreurs



Anatomie d'un QR Code

Lisibilité

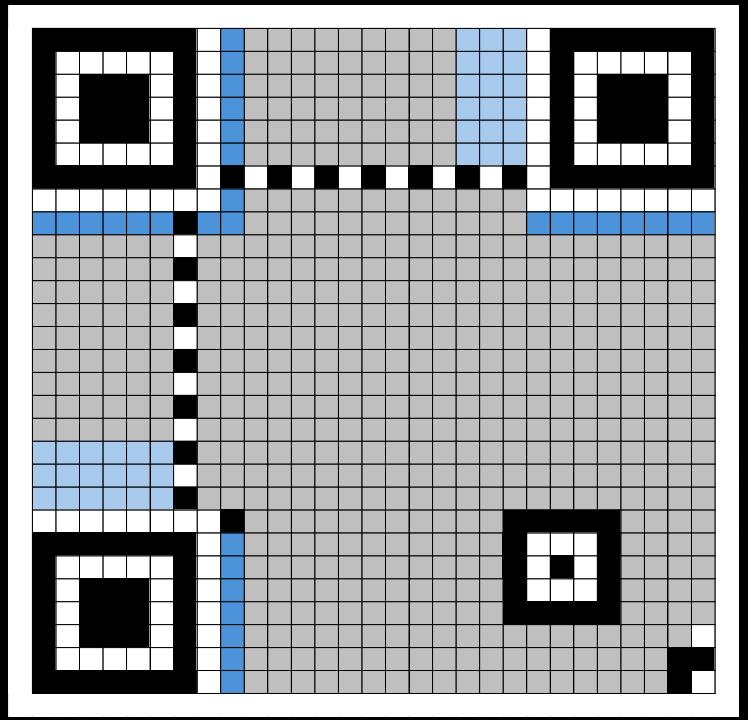
- Identification et focus
  - *Quiet zone*
  - 3 marqueurs de position
- *Alignment Patterns* pour gérer l'orientation

Modules

- Contenu stocké au format binaire
- Carré noir = 1 | Carré blanc = 0
- *Timing pattern* pour fixer la taille des modules

Format : *Level of Error Correction* et *Mask pattern*

Version



v3: 29x29

Correction d'erreurs

Données supplémentaires

- Nouvelles lignes et colonnes
- Augmente la taille du QR Code
- Compensent les données manquantes

Implémentation

- *Error Correction Codes* de Reed-Solomon
  - Polynômes et corps finis
  - Utilisé pour le stockage (DVD, RAID6) et le réseau (DSL, sat)
- 8-bit codewords pour les données et codes de correction

4 niveaux de correction

Utilisé pour injecter des images (vues comme des erreurs)



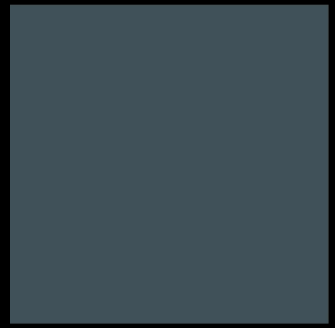
Niveau	% Erreur
Low	~7%
Medium	~15%
Quartile	~25%
High	~30%

IA Générative

Génération d’images

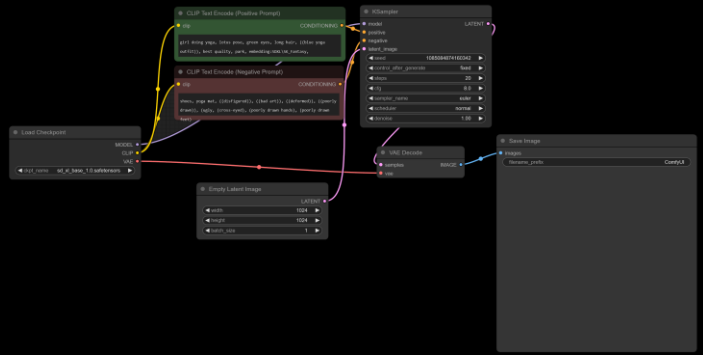
Stable Diffusion

- Débruitage progressif d’une image latente
- Guidé par le *prompt* (T2I) et l’image (I2I)
- Modèles « ouverts » et communautés dynamiques



ComfyUI

- IHM modulaire pour les flux de génération
- Exécution locale (Windows, Mac, Linux)
- Licence GPL 3



ControlNets

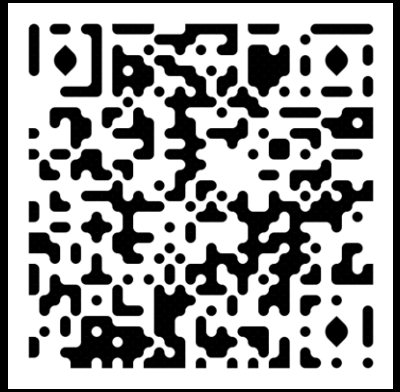
Réseaux de neurones spécialisés

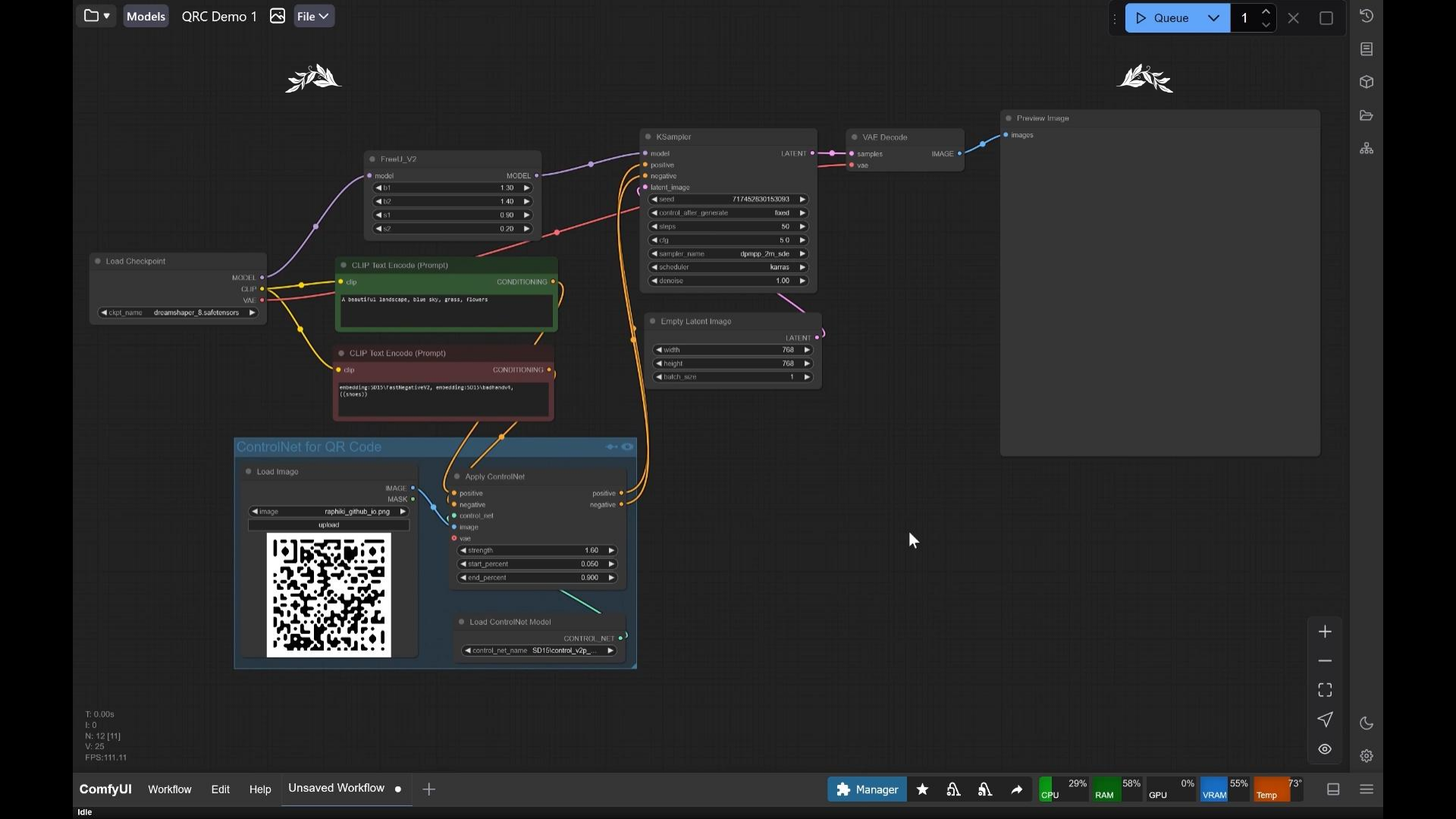
- Ajoutent plus de contrôle et de guidance
- A partir d'uniquement certaines caractéristiques d'une image

→ Approche hybride avec références textuelles et visuelles

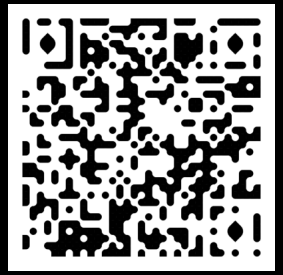
QR Code Monster

- "*This model is made to generate creative QR codes that still scan*"
- Essayons-le !




Models
QRC Demo 1
File
Queue
1
FreeU_V2
model
MODEL
b1 1.30
b2 1.40
s1 0.90
s2 0.20
KSampler
model
positive
negative
latent_image
LATENT
seed 717452830153093
control_after_generate fixed
steps 50
cfg 5.0
sampler_name dpmpp_2m_sde
scheduler karras
denoise 1.00
VAE Decode
samples
vae
IMAGE
Preview Image
images
Load Checkpoint
MODEL
CLIP
VAE
ckpt_name dreamshaper_8.safetensors
CLIP Text Encode (Prompt)
clip
CONDITIONING
A beautiful landscape, blue sky, grass, flowers
CLIP Text Encode (Prompt)
clip
CONDITIONING
embedding:SD15\FastNegativeV2, embedding:SD15\badhandv4, ((shoes))
Empty Latent Image
LATENT
width 768
height 768
batch_size 1
ControlNet for QR Code
Load Image
IMAGE
MASK
image raphiki_github_io.png
upload
Apply ControlNet
positive
negative
control_net
image
vae
strength 1.60
start_percent 0.050
end_percent 0.900
positive
negative
Load ControlNet Model
CONTROL_NET
control_net_name SD15\control_v2p_...
T: 0.00s
I: 0
N: 12 [11]
V: 25
FPS:111.11
ComfyUI
Workflow
Edit
Help
Unsaved Workflow
Manager
CPU 29%
RAM 58%
GPU 0%
VRAM 55%
Temp 73°
Idle

Modifier un QR Code via le *prompt*

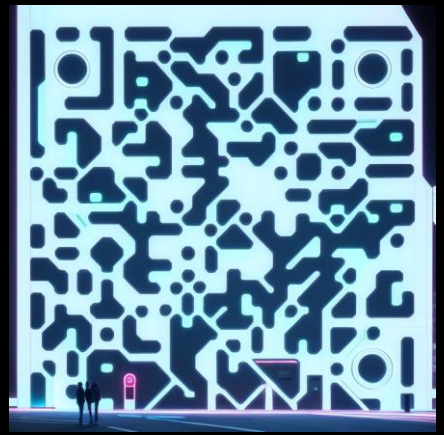


+

A beautiful landscape, blue sky, grass, flowers

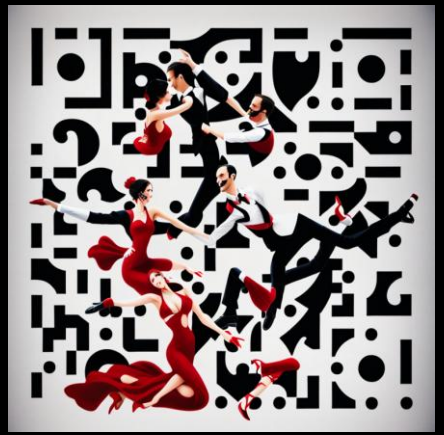


Modifier un QR Code via le *prompt*



A futuristic building with neon glows

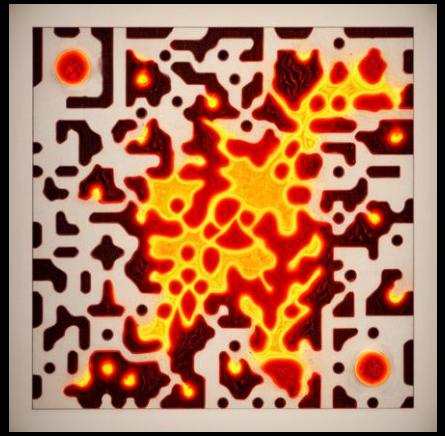
→ Style



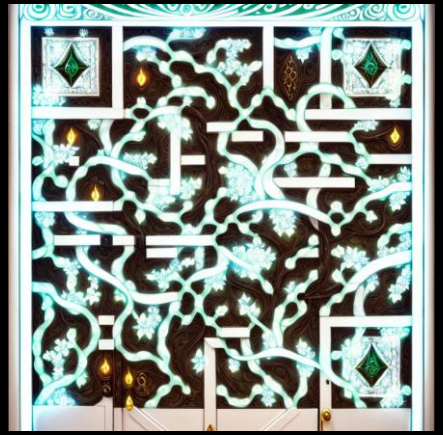
A couple dancing tango, woman with red dress, man in black tuxedo

→ Contenu

Via un *prompt* généré avec un LLM



A pattern forged from molten lava, glowing with an intense fiery orange and red hue. Cracks in the surface reveal a volcanic heat, with small embers rising around it"



An elegant, glowing elven door adorned with intricate, nature-inspired patterns and shimmering silver runes. Delicate vines and luminescent flowers intertwine with the carvings, pulsating with soft emerald and sapphire light. The archway, crafted from ethereal white stone, radiates a mystical aura, with faint golden mist swirling at its base, hinting at an ancient portal to a hidden realm

Modifier le style via le *prompt*



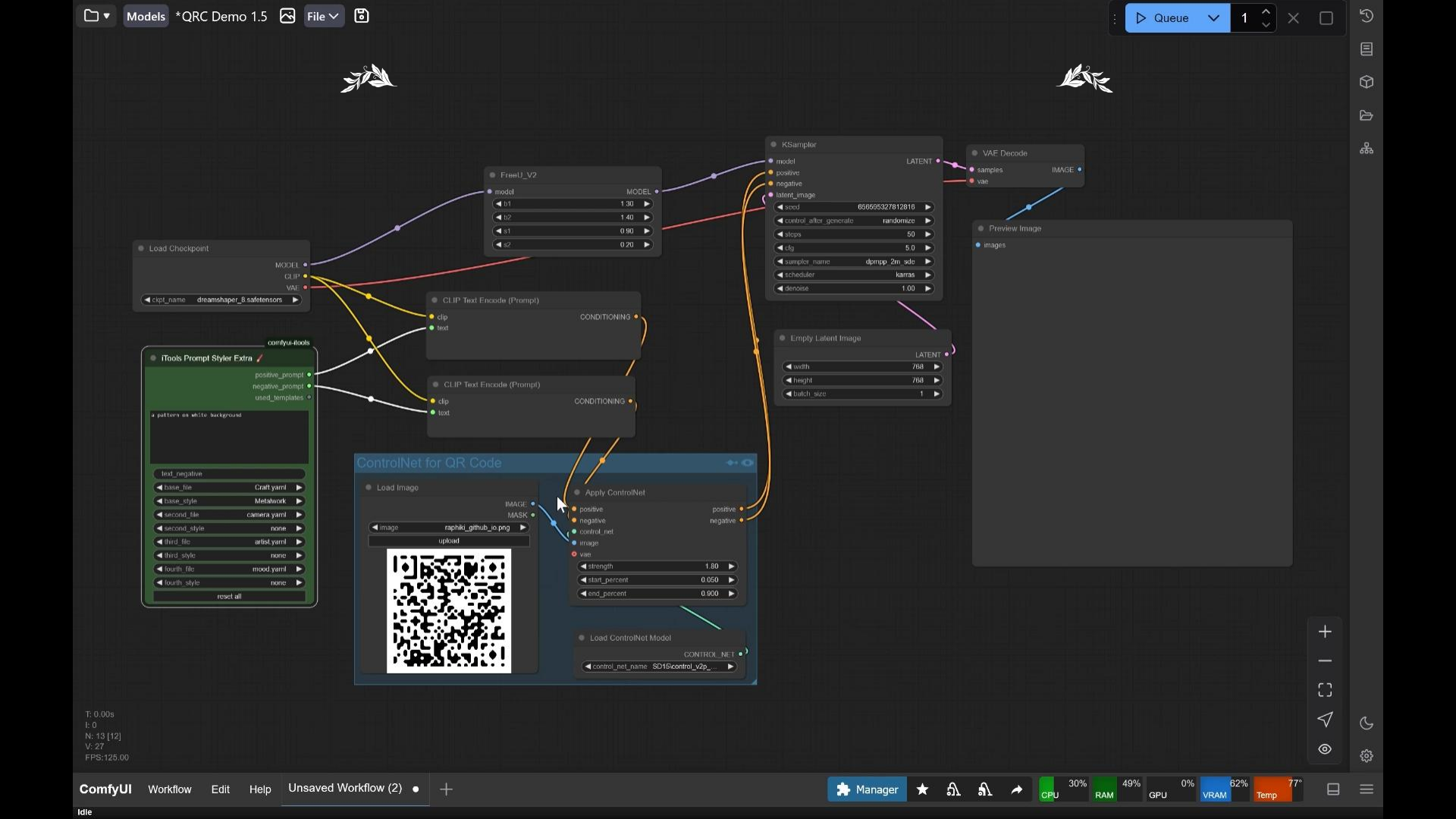

```yaml
- name: Macrame
- name: Marquetry
  negative_prompt: signature, watermark, ugly, off-center, deformed, cropped, worst quality, low quality, lowres, JPEG artifacts
  prompt: Marquetry art style of {prompt}, intricate wood inlay technique, creates decorative patterns and images using veneers of different woods, showcases craftsmanship and attention to detail

- name: Metalwork
  negative_prompt: signature, watermark, ugly, off-center, deformed, cropped, worst quality, low quality, lowres, JPEG artifacts
  prompt: Metalwork art style of {prompt}, manipulation of metal to create sculptures, showcases craftsmanship and technical skill, intricate and durable designs

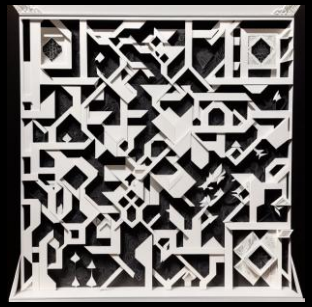
- name: Mosaic
  negative_prompt: worst quality, blurry, normal quality, signature, watermark, ugly, off-center, deformed, cropped, worst quality, low quality, lowres, JPEG artifacts, low details, bad photo, bad photography, bad art
  prompt: mosaic art style of {prompt}, versatile and decorative, uses small pieces of tile, adds color and texture, glossy, sharp
```

Ajout d'un *Custom Node* dans ComfyUI

- Elements de *prompt* réutilisable pour orienter le style
- De nombreuses catégories : 3D, Art, Artisanat, Design, Dessin, Illustration, Peinture, Sculpture, Vecteur...


Models
*QRC Demo 1.5
File
Queue
1
KSampler
model
positive
negative
latent_image
LATENT
seed 656595327812816
control_after_generate randomize
steps 50
cfg 5.0
sampler_name dpmpp_2m_sde
scheduler karras
denoise 1.00
VAE Decode
samples
vae
IMAGE
Preview Image
images
FreeU_V2
model
MODEL
b1 1.30
b2 1.40
s1 0.90
s2 0.20
Load Checkpoint
MODEL
CLIP
VAE
ckpt_name dreamshaper_8.safetensors
CLIP Text Encode (Prompt)
clip
text
CONDITIONING
CLIP Text Encode (Prompt)
clip
text
CONDITIONING
comfyui-itools
iTools Prompt Styler Extra
positive_prompt
negative_prompt
used_templates
a pattern on white background
text_negative
base_file Craft.yaml
base_style Metalwork
second_file camera.yaml
second_style none
third_file artist.yaml
third_style none
fourth_file mood.yaml
fourth_style none
reset all
Empty Latent Image
LATENT
width 768
height 768
batch_size 1
ControlNet for QR Code
Load Image
IMAGE
MASK
image raphiki_github_io.png
upload
Apply ControlNet
positive
negative
control_net
image
vae
strength 1.80
start_percent 0.050
end_percent 0.900
Load ControlNet Model
CONTROL_NET
control_net_name SD15\control_v2p_...
T: 0.00s
I: 0
N: 13 [12]
V: 27
FPS:125.00
ComfyUI
Workflow
Edit
Help
Unsaved Workflow (2)
Manager
CPU 30%
RAM 49%
GPU 0%
VRAM 62%
Temp 77°
Idle

Modifier le style via le *prompt*



Craft | Shadowbox > Paper



Craft | Macrame



3D | Abstract > Fluid Dynamics



Art | African Traditional



Art | Rococo



Experimental | Quantum Sketching



Basic | Expressionist



Basic | Gothic

Mieux contrôler l’image insérée

ControlNets supplémentaires

- Transfert de la posture via détection de la profondeur (*depth*) et des bords (*edges*)
- *Open Pose* n’est pas assez précis pour les postures de Yoga

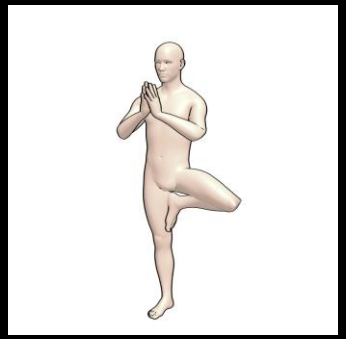


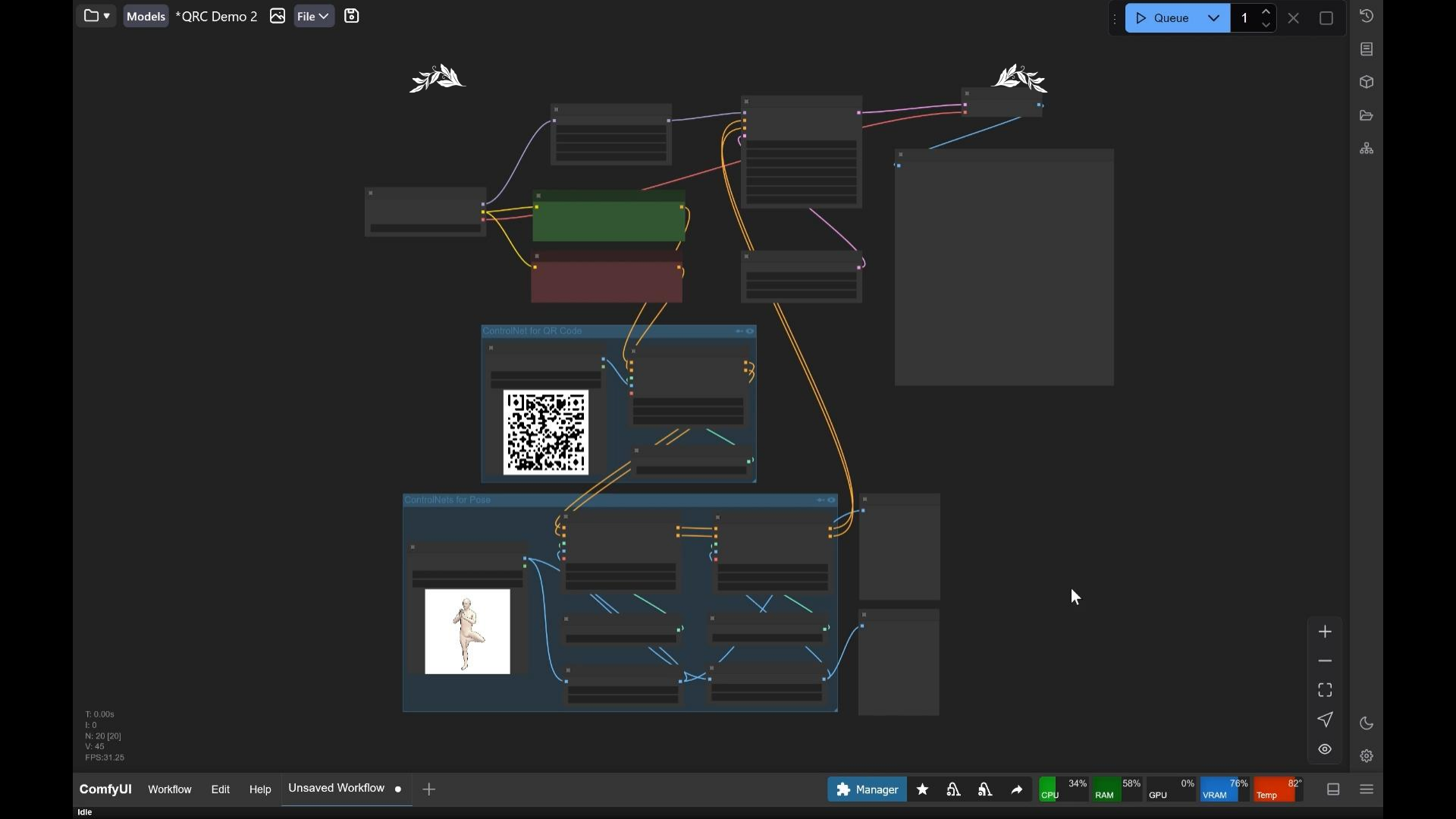
Image d’origine



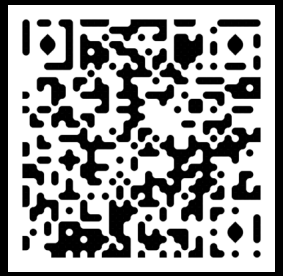
Depth



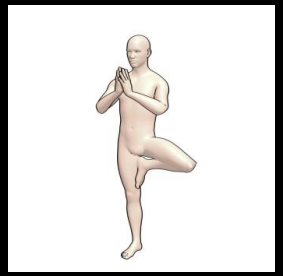
Canny Edge


Models
*QRC Demo 2
File
Queue
1
ControlNet for QR Code
ControlNets for Pose
T: 0.00s
I: 0
N: 20 [20]
V: 45
FPS:31.25
ComfyUI
Workflow
Edit
Help
Unsaved Workflow
Manager
CPU 34%
RAM 58%
GPU 0%
VRAM 76%
Temp 82°
Idle

Transférer une posture



+



+

1 man, mixed race, short curly hair, black hair, 40 years old, white t-shirt, black yoga pants, short sleeves, smiling, viewing glasses, white background, barefoot



Transférer un visage

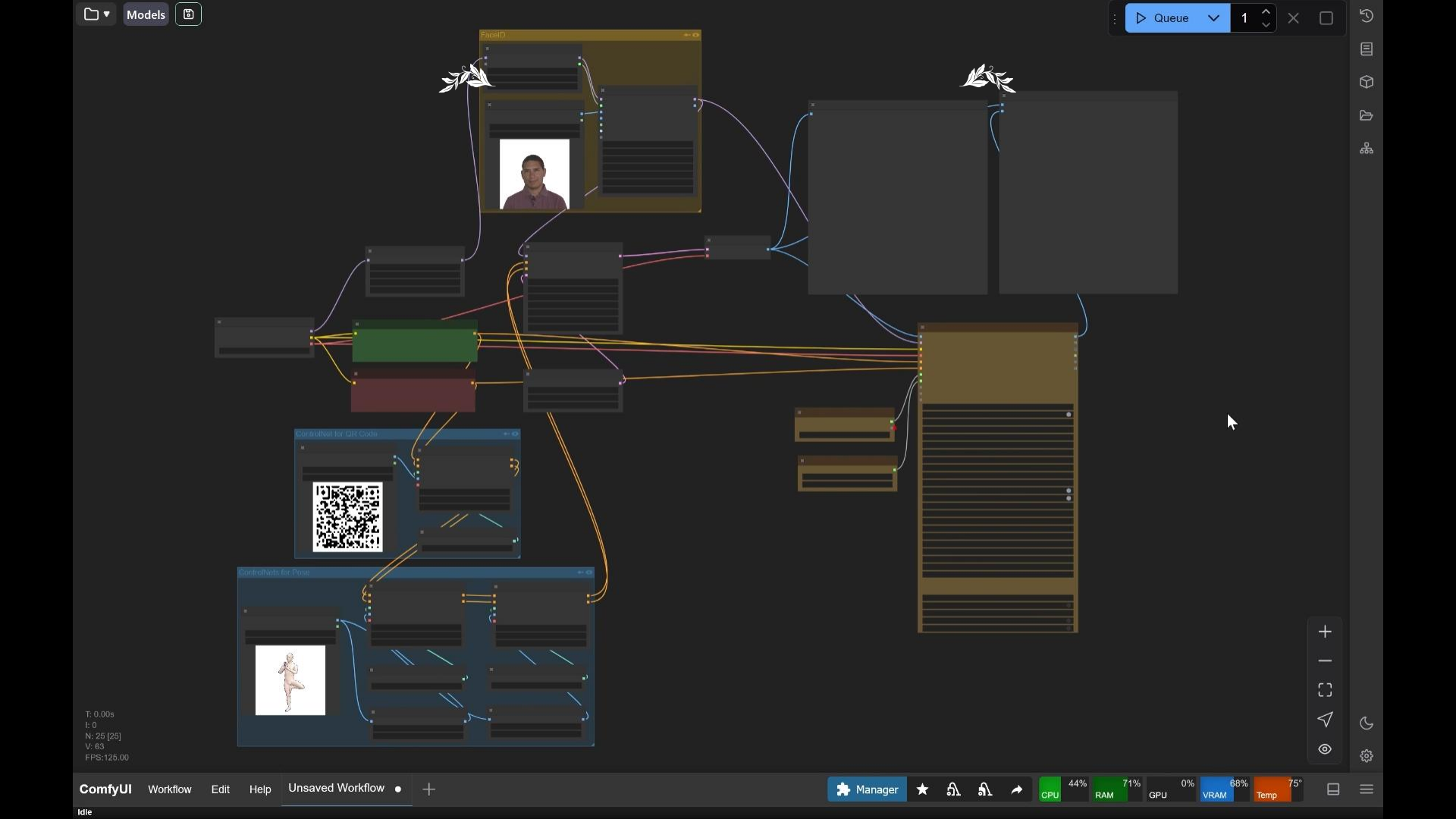
Image Prompt Adapter

- Utiliser une image existante comme *prompt*
- Modèles pré-entraînés supplémentaires
- FaceID : modèle de reconnaissance faciale

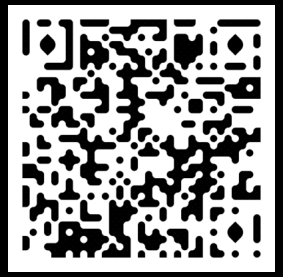
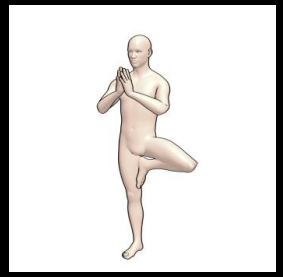


FaceDetailer

- Modèle de traitement d'images post-génération
- Amélioration du visage (yeux, nez, lèvres, expression…)


Models
Queue
1
FaceID
ControlNet for QR Code
ControlNets for Pose
T: 0.00s
I: 0
N: 25 [25]
V: 63
FPS:125.00
ComfyUI
Workflow
Edit
Help
Unsaved Workflow
Manager
CPU 44%
RAM 71%
GPU 0%
VRAM 68%
Temp 75°
Idle

Transférer un visage


+

+

+

1 man, mixed race, short curly hair, black hair, 40 years old, white t-shirt, black yoga pants, short sleeves, smiling, viewing glasses, white background, barefoot



Transfert de style, pose et visage



Craft | Gobelin Tapestry



Painting | Watercolor > Cartoon

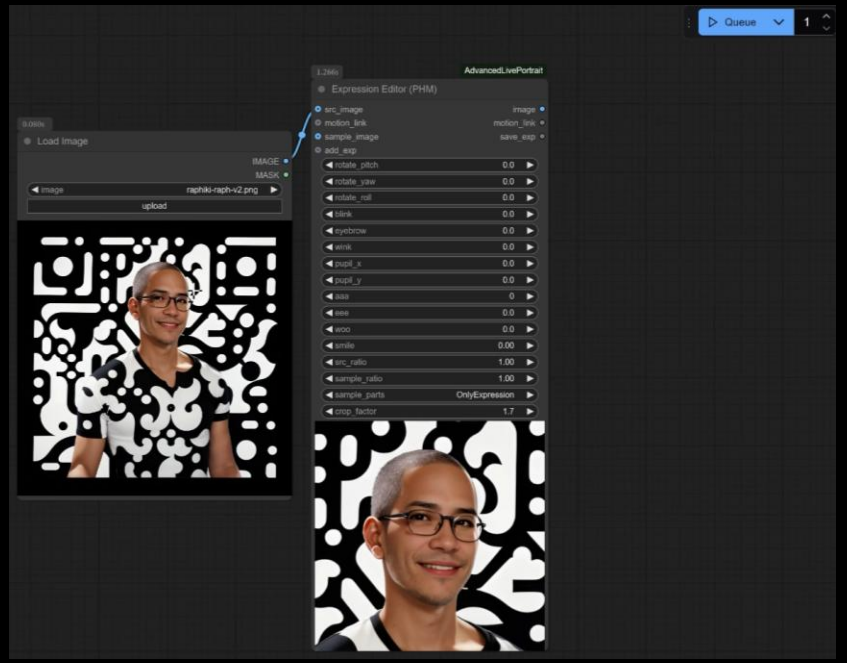


3D | Abstract > Fluid Dynamics

Animer un visage

Advanced Live Portrait

- Extrait les expressions faciales
- Modifie et insère des expressions
- Animation à l'aide de plusieurs expressions



Models

Queue 1

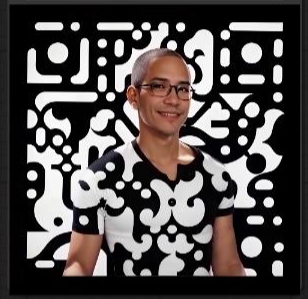
AdvancedLivePortrait

AdvancedLivePortrait

VideoHelperSuite

Load Image

IMAGE
MASK
image raphiki-raph-v2....
upload



Expression Editor (PHM)

src_image → image
motion_link → motion_link
sample_image → save_exp
add_exp

rotate_pitch	4.0
rotate_yaw	0.0
rotate_roll	2.0
blink	4.0
eyebrow	6.0
wink	0.0
pupil_x	0.0
pupil_y	0.0
aaa	5
eee	0.0
woo	0.0
smile	1.30
src_ratio	1.00
sample_ratio	1.00
sample_parts	OnlyExp...
crop_factor	1.7

Advanced Live Portrait (PHM)

src_images → images
motion_link
driving_images

retargeting_eyes	0.00
retargeting_mouth	0.00
crop_factor	1.7
turn_on	true
tracking_src_vid	false
animate_without_vid	true

```
1 = 5:5
0 = 3:5
```

Video Combine

images → Filenames
audio
meta_batch
vae

frame_rate	15
loop_count	0
filename_prefix	AdvancedLivePortrait
format	image/gif
pingpong	false
save_output	true

Note

orginal src : motion 0

Note

```
Command Description
1 = 5:5
0 = 3:5

[Motion index] = [Changing frame length] : [Length of frames waiting for next motion]

for example, first line -> 1 = 5:5
Change over 5 frames with motion 1, and wait for the next motion for 5 frames
```

Note

motion 1

T: 0.00s
I: 0
N: 7 [7]
V: 14
FPS:23.26

ComfyUI | Workflow | Edit | Help | Unsaved Workflow (3) | Unsaved Workflow (4)

Manager | CPU 29% | RAM 59% | GPU 0% | VRAM 39% | Temp 76°

Idle

Animer un visage



Merci



Celui-là scannez le !